Title

Geospatial Information, Metadata, and Maps for Global River Corridor Science Focus Area Sites (v5)

Summary

This dataset provides geospatial information, metadata, and maps for the Pacific Northwest National Laboratory (PNNL) River Corridor Science Focus Area (RC-SFA; https://www.pnnl.gov/projects/river-corridor) sites. The RC-SFA works to transform understanding of spatial and temporal dynamics in river corridor hydrobiogeochemical functions from molecular reaction to watershed and basin scales. The knowledge we gain is used to formulate and test hypotheses and to improve mechanistic representation of river corridor processes and their response to disturbances in multiscale models of integrated hydrobiogeochemical function. The data provided includes Site ID, latitude, longitude, stream name, and common ID (COMID) for sites used across the RC-SFA. The COMID can be used to find and download data from NHDPlus (https://www.epa.gov/waterdata/nhdplus-national-hydrography-dataset-plus) and other platforms. The sites included are non-exhaustive. Sites (including past sites) will be added to this data package in the future. Data generated from the RC SFA can be accessed at https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA.

This data package was originally published in April 2023. It was updated in June 2023 (v2; modified files), December 2023 (v3; modified files), January 2025 (v4; modified files), and December 2025 (v5; modified files). See the change history section below for more details.

Critical Details

1 – Locations for each site were collected by field teams using a variety of GPS technologies. Each site's location was matched with the nearest **NHDPlus Version 2** (NHDPlusV2; https://www.epa.gov/waterdata/get-nhdplus-national-hydrography-dataset-plus-data) network streamline using either the NNJoin nearest-neighbor spatial join plugin (https://plugins.qgis.org/plugins/NNJoin/) for QGIS (QGIS Geographic Information System. QGIS Association. http://www.qgis.org) or R (the R script can be found at https://github.com/river-corridors-sfa/Geospatial_variables/blob/71149dc0623a5a4c28458e215a3c06e01d0169c7/Example_Code/geospatial_workflow_check_comids.Rmd). Streamlines are identified in NHDPlusV2 by a unique common-ID or COMID. The COMID identified by the nearest-neighbor spatial join is referred to here as the raw COMID. Join results (raw COMIDs) were manually reviewed by personnel familiar with the sites. One key review criterion was whether or not the stream name from the site metadata matched the stream name from NHPPlusV2. In situations where the nearest-neighbor spatial join did not identify the correct stream, the correct stream line was manually identified (if possible), and a rectified COMID was extracted.

Data Package Structure

This dataset is comprised of one main data folder. The data folder consists of (1) file-level metadata; (2) data dictionary; (3) readme; (4) methods codes; (5) geospatial information for all RC SFA sites including International Generic Sample Number (IGSN); (6) maps of all sites and sites in Washington State, USA; and (7) a subfolder with the shapefile of all sites. All files are .csv, .pdf, .shp, .cpg, .dbf, .prj, .qmd, or .shx.

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Change History

Approach to change history and versioning:

Updates to **data package** version: When any file within a data package is updated, the data package version number is updated. The data package version number is indicated in the title of the data package, the data package folder name, and in the change history table below. You can access previous versions of the data package by sending a request to ESS-DIVE.

Updates to **individual file** versions: As files are changed, the file version number is also updated. The file version number is indicated in the file name, file level metadata (flmd) file, and the change history table below. The version number on an individual file may not match the version number of the data package. For example, v3 of a data package may include v2 of an individual file.

The change history below describes each file revised during versioning. If you are interested in seeing the exact cells within a file that have changed, you can utilize the daff package in R (https://github.com/edwindj/daff) to compare a previously downloaded file to a newly downloaded file.

In the change history table below, the sub-headers and bullets indicate the type of change in each file:

- New files: Describes new files added that were not present in previous data package versions.
- Bulk changes to files: Describes a change to many files within the data package. The indicated superscript will be added to each file name that the change applies to.
- Modified files:
    - Corrected: Describes existing information modified or removed to prevent sharing of incorrect information
    - Added: Describes new information inserted into an existing file (e.g., appending new columns/rows)
    - Updated: Describes modifying existing information to maintain accuracy though version changes. (e.g., changing version number to new version number)

Change history:

Data Package Version	Changes
Version 1 *April 2023*	Original data package publication.

Version 2 *June 2023*	**MODIFIED FILES** **RCSFA_Geospatial_Site_Information.csv** (v2) • Corrected coordinates and COMID for site S47R. • Added sites S55N and S56N to distinguish the sites that were visited in 2022 (S55N and S56N) from the sites visited in 2021 (S55 and S56). **RCSFA_Geospatial_Sites_*_Map.pdf** (v2) • Corrected coordinates for site S47R. • Added new sites to the map. **RCSFA_Geospatial_Sites_Shapefile.*** (v2) • Corrected coordinates for site S47R. • Added new sites to the shapefile. **readme_RCSFA_Geospatial.pdf** (v2) • Added to change history table to reflect new changes made in this version. • Updated version number in data package title. **RCSFA_Geospatial_flmd.csv** (v2) • Updated version number on file names and file paths to reflect recent changes.
Version 3 *December 2023*	**MODIFIED FILES** **RCSFA_Geospatial_Methods_Codes.csv** (v2) • Corrected the method description for COMID_06 to make it more inclusive. **RCSFA_Geospatial_Site_Information.csv** (v3) • Corrected 25 Site IDs to maintain consistency with updates made to other data packages. • Corrected the "Methods_and_Flags" column for 2 rows (sites E431 and U22). • Added 61 new rows to include new sites. There are now 458 rows. • Added new methods codes to all rows to describe the software used to identify the COMIDs. **RCSFA_Geospatial_dd.csv** (v2) • Added new row to define the "Data_Orientation" column in the flmd. • Updated rows to reflect changes in column headers of modified files. • Updated definition of IGSN from "International Geo Sample Number" to "International Generic Sample Number" to reflect the new name change. • Updated the "Missing_Value_Codes" definition to better describe the column. **RCSFA_Geospatial_flmd.csv** (v3) • Added new column called "Data_Orientation". • Updated text used to describe the flmd file. • Updated version number on file names and file paths to reflect recent changes.

	• Updated column headers from “Start_Date” and “End_Date” to “Date_Start” and “Date_End”. • Updated “Date_End” value to “2023-10-10” to reflect the newly added data. **RCSFA_Geospatial_Sites_*_Map.pdf** (v3) • Added new sites to the map. **RCSFA_Geospatial_Sites_Shapefile.*** (v3) • Added new sites to the shapefile. **readme_RCSFA_Geospatial.pdf** (v3) • Added version information to Summary section. • Added new versioning information, including the text to describe versioning approach and changing the formatting of the change history table. • Updated the Critical Details section to better describe how the COMIDs were identified. • Updated definition of IGSN from “International Geo Sample Number” to “International Generic Sample Number” to reflect the new name change. • Updated version number in data package title.
Version 4 *January 2025*	**MODIFIED FILES** **RCSFA_Geospatial_Site_Information.csv** (v4) • Added 5 new rows to include new sites. **RCSFA_Geospatial_flmd.csv** (v4) • Updated version number to modified files. **RCSFA_Geospatial_Sites_*_Map.pdf** (v4) • Added new sites to the map. **RCSFA_Geospatial_Sites_Shapefile.*** (v4) • Added new sites to the shapefile. **readme_RCSFA_Geospatial.pdf** (v4) • Updated new versioning information. • Updated version number in data package title.
Version 5 *December 2025*	**MODIFIED FILES** **RCSFA_Geospatial_Site_Information.csv** (v5) • Added new rows to include additional sites. • Added “Origin_Study_Code” column. • Removed rows with redundant sites from S19S. **RCSFA_Geospatial_Methods_Codes.csv** (v3) • Added data flag “Flag_01”. **RCSFA_Geospatial_flmd.csv** (v5) • Updated version number for modified files. • Updated columns to reflect most recent ESS-DIVE guidance. **RCSFA_Geospatial_dd.csv** (v3) • Updated to reflect columns in modified files.

	• Updated columns to reflect most recent ESS-DIVE guidance. **RCSFA_Geospatial_Sites_*_Map.pdf** (v5) • Added new sites to the map. **RCSFA_Geospatial_Sites_Shapefile.*** (v5) • Added new sites to the shapefile. **readme_RCSFA_Geospatial.pdf** (v5) • Updated new versioning information. • Updated version number in data package title.

Note: An asterisks (*) in a file name indicates multiple files with the same file and file naming structure. See the flmd for more details about specific files and the meaning of the asterisks.